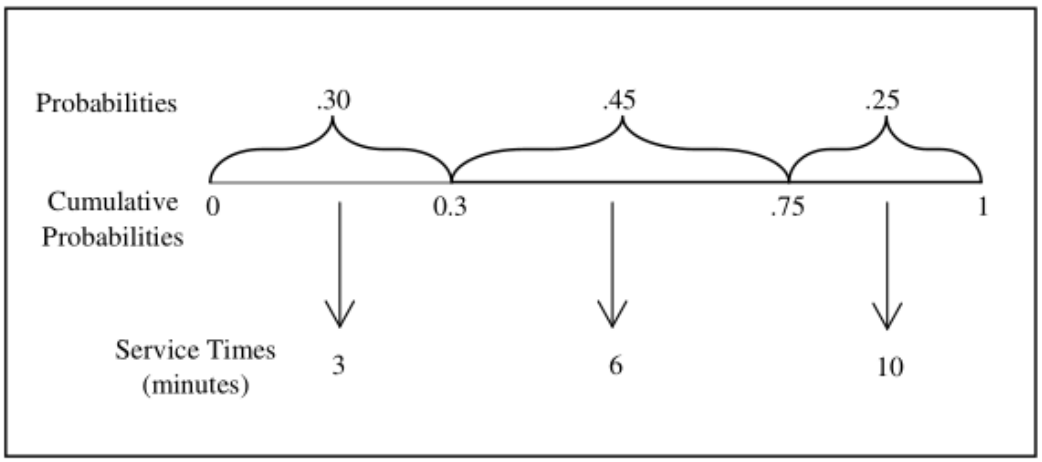
How to Simulate Random Service Times using Excel

- In this example we introduce a method for generating random samples from a discrete distribution, an essential skill in simulation modelling. This method will help us in accurately modelling various real-world systems, such as queues, inventories, and more.
- For our example we have an automated calling station that spends either 3, 6 or 10 minutes on each customer.
- **Input Specification:** We start by defining the probabilities for each service time (3, 6, or 10 minutes) and their cumulative probabilities. The cumulative probabilities always add up to 1.0.
- The probability of each call length is 0.30, 0.45, and 0.25 respectively. Therefore the cumulative probabilities are 0.3, (0.3+0.45), (0.75+0.25).

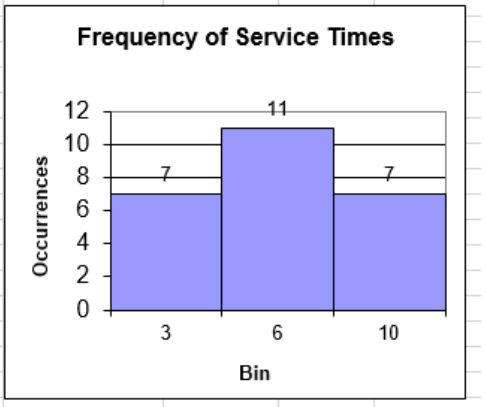
Service Time	Probability	Cumulative Probability
3	0.30	0.30
6	0.45	0.75
10	0.25	1.00

- We use random numbers to determine the service time. Think of it like throwing a dart at a dartboard. The position where the dart lands on a line between 0 and 1 determines the service time.
- Transformation is the key principle in the process of generating random service times. It means turning a random number into a service time of a prescribed probability.
- To transform, we compare the random number to the cumulative probabilities to determine the service time. For example, if the random number is less than or equal to 0.30, the service time is 3 minutes. If it's between 0.30 and 0.75, the service time is 6 minutes. If it's greater than 0.75, the service time is 10 minutes.



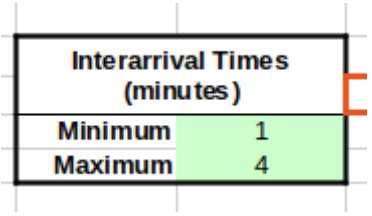
- For example, we can generate 5 random probabilities using *RAND or Rnd1* in excel as follows: 0.9871, 0.0226, 0.0008, 0.2128, 0.8586.
- After transformation, we end up with random service times 10, 3, 3, 3, 10.
- We can apply the logic of transformation in Excel using the Visual Basic function DiscreteEmp() (for discrete empirical, meaning defined by data)
- Simulation Table: We use these methods to generate a simulation table that shows the resulting frequency distribution (or histogram) of the generated service times for a specified number of callers.
- The full simulation table can be found at Example2.5Service Times.xls

Number of Callers =	25
Simulation Table	
Step	Activity
Caller	Service Time
1	6
2	6
3	10
4	6
5	6
6	6
7	10
8	3
9	3
10	3
11	10
12	6
13	6
14	10
15	3
16	10
17	6
18	10
19	6
20	3
21	10
22	6
23	3
24	6
25	3



How to Simulate Random Arrival Times using Excel

- Interarrival times, like the service times in the previous example, are essential components of simulation models, representing the durations of various activities within the system.
- To understand these concepts better, let's first clarify some terminology. We'll encounter two types of times in our simulation: *activity times* and *arrival times*. Activity times, such as interarrival times and service times, represent the duration of specific actions within the system. On the other hand, arrival times denote the moments when customers or entities arrive at the system.
- In our simulations, we use a simulation CLOCK to keep track of time. This CLOCK helps us record when events occur, such as arrivals or service completions, allowing us to observe how the system evolves over time.
- In our Excel spreadsheets, we'll label the simulation CLOCK times above the relevant columns to distinguish them from the activity times.
- For this example, we'll focus on generating random arrival times for telephone calls to our information service.
- These arrivals will occur at random intervals, following a discrete distribution where interarrival times can be 1, 2, 3, or 4 minutes, each with an equal probability.
- To generate these interarrival times randomly, we use the VB function, `DiscreteUniform()`, specifically designed for generating random values from a discrete uniform distribution (all possible interarrival times have equal probabilities of occurring).
- The `DiscreteUniform()` function takes two arguments: the minimum and maximum values of the desired range of integers, in this case we pass the cell addresses containing the lowest and highest values as follows:



Interarrival Times (minutes)	
Minimum	1
Maximum	4

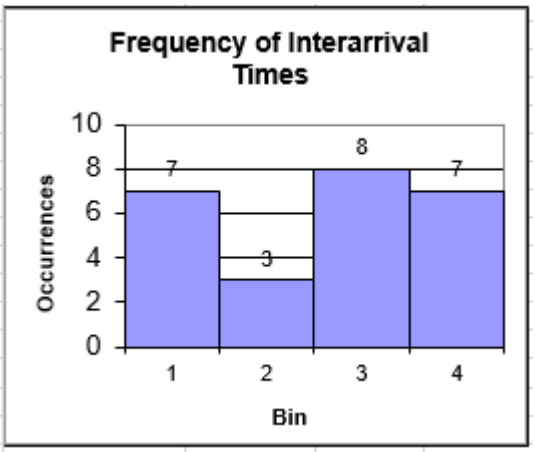
```
DiscreteUniform($G$5,$G$6)
```

- In our spreadsheet model, we generate the interarrival times for all but the first customer (whose arrival time is assumed to be 0).
- We can then calculate the arrival times for each customer by simply adding each customer's inter-arrival time to the previous customer's arrival time.

	A	B	C	D
4		**Interarrival Times (minutes)**		
5				
6		**Minimum**	1	
7		**Maximum**	4	
8				
9		**Number of Callers=**		25
10		**Simulation Table**		
11		**Step**	**Activity**	**Clock**
12			**Interarrival**	
13		**Caller**	**Time**	**Arrival Time**
14		1		0
15		2	2	2
16		3	1	3
17		4	3	6

- Simulation Table: We use these methods to generate a simulation table that shows the resulting frequency distribution (or histogram) of the generated arrival times for a specified number of callers.
- Full simulation table can be found at [Example2.3ArrivalTimes.xls]

Number of Callers =		25
Simulation Table		
Step	Activity	Clock
Caller	Interarrival Time	Arrival Time
1	1	0
2	2	2
3	1	3
4	3	6
5	1	7
6	3	10
7	4	14
8	3	17
9	3	20
10	4	24
11	3	27
12	2	29
13	4	33
14	1	34
15	1	35
16	1	36
17	4	40
18	3	43
19	3	46
20	4	50
21	1	51
22	4	55
23	2	57
24	4	61
25	3	64



References

Banks, Carson II, J., Nelson, J., Nicol, B. (2014). Simulation examples in spreadsheet. In *Discrete-Event System Simulation* (pp. 30-34). Pearson.